Solution

(i) In this equation, the highest order derivative is $\frac{dy}{dx}$ whose power is 1

Therefore, the given differential equation is of order 1 and degree 1.

(ii) Here, the highest order derivative is $\frac{d^4y}{dx^4}$ whose power is 3.

Therefore, the given differential equation is of order 4 and degree 3.

(iii) In the given differential equation, the highest order derivative is $\frac{d^2y}{dx^2}$ whose power is 1.

Therefore, the given differential equation is of order 2.

The given differential equation is not a polynomial equation in its derivatives and so its degree is not defined.

(iv) The given differential equation is $3\left(\frac{d^2y}{dx^2}\right)=\left[4+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}$

Squaring both sides, we get $9\left(\frac{d^2y}{dx^2}\right)^2=\left[4+\left(\frac{dy}{dx}\right)^2\right]^3$.

In this equation, the highest order derivative is $\frac{d^2y}{dx^2}$ whose power is 2.

Therefore, the given differential equation is of order 2 and degree 2.

(v) $dy+(xy-\cos x)dx=0$ is a first order differential equation with degree 1, since the equation can be rewritten as $\frac{dy}{dx}+xy-\cos x=0$.

EXERCISE 10.1

1. For each of the following differential equations, determine its order, degree (if exists)

(i) $\frac{dy}{dx}+xy=\cot x$ (ii) $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}}-3\frac{d^2y}{dx^2}+5\frac{dy}{dx}+4=0$

(iii) $\left(\frac{d^2y}{dx^2}\right)^2+\left(\frac{dy}{dx}\right)^2=x\sin\left(\frac{d^2y}{dx^2}\right)$ (iv) $\sqrt{\frac{dy}{dx}}-4\frac{dy}{dx}-7x=0$

(v) $y\left(\frac{dy}{dx}\right)=\frac{x}{\left(\frac{dy}{dx}\right)+\left(\frac{dy}{dx}\right)^3}$ (vi) $x^2\frac{d^2y}{dx^2}+\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{1}{2}}=0$

(vii) $\left(\frac{d^2y}{dx^2}\right)^3=\sqrt{1+\left(\frac{dy}{dx}\right)}$ (viii) $\frac{d^2y}{dx^2}=xy+\cos\left(\frac{dy}{dx}\right)$

(ix) $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+\int ydx=x^3$ (x) $x=e^{xy\left(\frac{dy}{dx}\right)}$

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